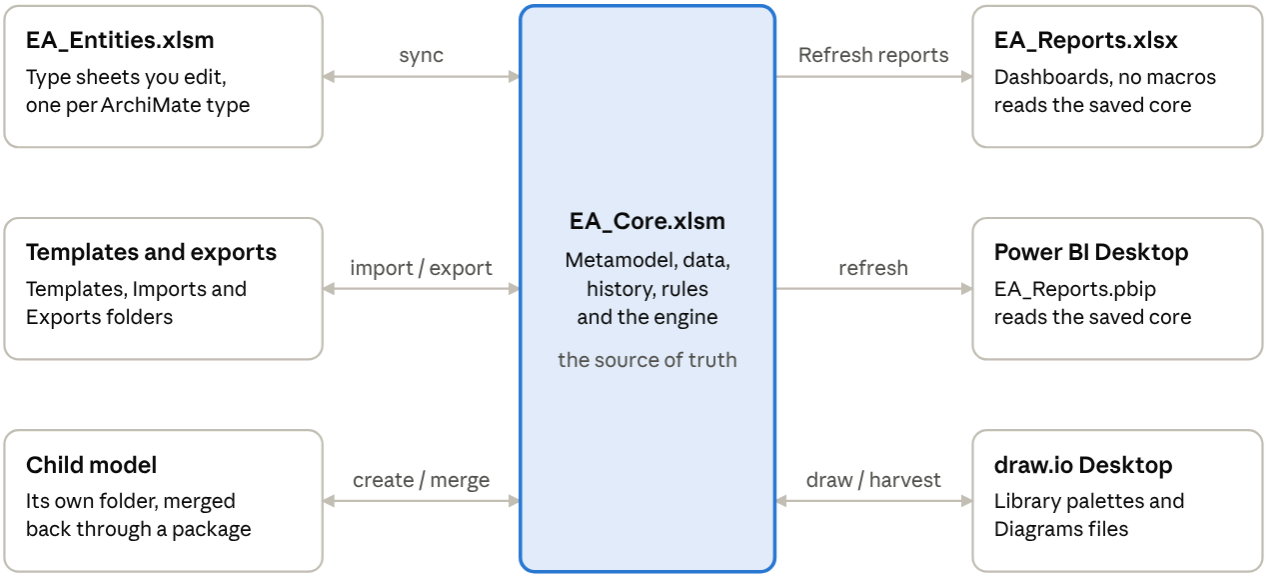
EA Repository Workbench: User Guide

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At a glance

The Workbench keeps one model per folder. You edit in EA_Entities; EA_Core holds the metamodel, the data and the history, and every other tool reads or writes it through files in the same folder.

EA_Core holds the model; the other tools reach it through files



the Workbench at a glance · one core, six connections

EA_Reports and Power BI read the last saved EA_Core, so save (or use Refresh reports) before refreshing them.

Daily work

You work in **EA_Entities**: open it, edit on the type sheets, save. Everything else happens behind the scenes in EA_Core.

Open the model. Open EA_Entities.xlsm. It opens EA_Core for you and runs a quick sync; when you close EA_Entities, EA_Core closes too. Each model is its own folder, and only one model can be open at a time.

Find your way. Home lists every type sheet (T_ApplicationComponent, T_Capability and so on) with its entity count. Click a name to jump to it.

Edit entities. Each row is an entity, each column an attribute that applies to its type; required columns are marked with *. Type over a value to change it. To add an entity, type on a blank row below the data: the Workbench gives it a Display ID such as APP-0004 when you save.

See relationships. Click the **+** in the left margin of a row to open its relationship rows; click a related name to jump to that entity.

Save. Saving EA_Entities sends your edits to EA_Core, with history. Rows with errors stay highlighted and queued until they are fixed.

Remove an entity. Set its Status to **Retired**. Deleting the row does not delete anything; the next sync puts it back.

Right-click a row on a type sheet for:

Menu item	What it does
Edit this entity	The entity form, with every attribute and its rules
Add relationship from this entity	A form that offers only the relationships ArchiMate allows from this type
History of this entity	Every change to it and its relationships, newest first
Detail of selected entities	Attributes, relationships and recent changes on RPT_EntityDetail
Impact of this entity	Everything within 1 to 5 relationships on RPT_Impact
Diagram of selected entities	A draw.io diagram of the selection (see Diagrams)

The same forms are on Home: **Add entity (form)** and **Add relationship**.

To change a relationship's label or retire it, click + beside the entity, right-click the relationship row and choose **Change or retire this relationship**. The detailed User Guide covers every feature step by step.

Validation and rules

Every value is checked as you type: **red** is an error and blocks the save of that row, **amber** is a warning that saves. Hover over the cell to read the message.

Three layers of checks apply, in this order:

1. **Applicability:** a value in an attribute that does not apply to the entity's type is an error (those cells are shaded grey).
2. **Data type:** whole numbers, decimals, dates, yes/no, list values, URLs and e-mail addresses must look right.
3. **Rules** from MM_Rules: Mandatory, RequiredIf, List, MinValue, MaxValue, Range, MinLength, MaxLength, Regex, DateRange, DateOrder, Unique and EntityRef. Each rule has a severity, so the same check can be an error for one type and a warning for another.

Validate All (Home) re-checks the whole repository, including relationships against the allowed ArchiMate relationships, and lists every issue on SYS_Issues in EA_Core with a link to the cell. Sync Now runs it too.

Example: Application Component requires Owner and Lifecycle. A new row "Broker Portal" with Lifecycle Production and no Owner turns the Owner cell red with "Owner is required", and the row is not saved until an owner is filled in.

Changing the metamodel

The metamodel is data in EA_Core's MM_ sheets, seeded with ArchiMate 4's 42 elements and 11 relationships; change it with the wizards on EA_Core's Home, not by typing in the sheets. A snapshot is taken automatically before every metamodel change.

To	Use	Notes
Add an entity type	Add / edit type > Entity type	Name, domain, aspect and ID prefix (such as APP); a type sheet and a draw.io style follow
Add an attribute	Add / edit type > Attribute type	Data type (text, number, date, list and so on), list name, unit
Use an attribute on a type	Add / edit type, on the type	The mapping grid: tick the attribute, mark it required, set its order and section
Allow a relationship	Add / edit type > Relationship type (allowed grid)	Source type, relationship, target type; a group (G:NAME), * (any) or SAME (same type) also work
Add a rule	Define rule	Pick attribute and type, rule type, parameters and severity; Test checks a sample value first

To	Use	Notes
Retire a type or attribute	Set its Status to Retired	Never deleted, so history stays readable

After changes, click **Apply metamodel changes**: it raises the schema version, adds DATA_Master columns, and regenerates the validation and every type sheet. Then rebuild the draw.io stencil (Diagrams > Rebuild ArchiMate 4 stencil), or let the next Sync Now do it.

The allowed relationships are derived from the ArchiMate 4 generic metamodel and marked *ArchiMate4-derived (verify)* in MM_EntityRelationships. Check them against the relationship tables of your copy of the specification before relying on them.

Sync and SYNC_Review

EA_Core is the source of truth, and EA_Entities is kept in step with it automatically; you only step in when both sides changed the same thing.

When	What happens
Opening either workbook	Quick sync: queued edits are pushed, rows changed in EA_Core are repainted, sheet structure is fixed
Saving EA_Entities	Your edits are validated and written to EA_Core with history
Sync Now (both Homes)	Full sync: every row is compared, Validate All runs, the draw.io libraries and views register are refreshed
A workbook was saved without macros	Detected on the next open; those rows are compared in full

Differences that need a decision appear on **SYNC_Review** in EA_Core, one row per entity and attribute:

Situation	Default Keep
Edited in EA_Entities outside the Workbench	EA_Entities (apply the edit)
Changed in both EA_Core and EA_Entities	EA_Core
New row in EA_Entities	EA_Entities (add it)

Situation	Default Keep
Row in EA_Entities not found in EA_Core	EA_Core (remove it from the sheet)

Set Keep to EA_Core, EA_Entities or Skip (decide later), then click **Apply decisions**. A type sheet with open items is not repainted until they are settled, so nothing on it is overwritten.

Versions and history

Every change is recorded, and a snapshot of the whole model can be taken or restored at any time from **Version history** (both Homes).

- **Change history** is automatic: each H_ sheet in EA_Core logs who changed which attribute, old and new value, when, and from where (form, sheet, import, diagram, merge). History is never overwritten; past 100,000 rows, the oldest rows move to History\Archive_<year>.xlsx during Sync Now.
- **Take snapshot** saves a values-only copy of EA_Core and of the filled type sheets in History. One is taken automatically before metamodel changes, restores, imports, diagram harvests and merges.
- **Compare selected with current** or **Compare two versions** lists every added, removed and changed entity, relationship and metamodel row on VERSION_Compare.
- **Restore entity or type** puts one entity (by Display ID) or every entity of a type back to a version's values. **Restore whole model** replaces the metamodel, entities and relationships with a version. Restores are logged in history like any other change.
- **Entity history** (or right-click a row > History of this entity) lists every change to one entity and its relationships, newest first.

Example: someone overwrote Claims Platform's vendor last week. Open Version history, select last Monday's snapshot, click Restore entity or type, enter APP-0002, and confirm. Only that entity changes, and the restore shows in its history.

Templates, import and export

Bulk data comes in through templates and goes out through exports, both from **Templates, import, export** (EA_Core Home) or **Import / export** (EA_Entities Home).

Generate a template. Pick an entity type. You get a macro-free .xlsx in Templates\ that anyone can fill without the Workbench: an Entities sheet with dropdowns and required columns marked *, a Relationships sheet that offers only allowed relationships and target

types, and an Allowed sheet listing them. With **Pre-fill = Y** the existing entities are included for bulk updates; in a pre-filled template a cleared cell clears the value, in a fresh one blank cells are ignored.

Import. Put filled templates in Imports, then:

1. **Choose files and preview.** Rows match by Entity ID, then by exact name within the type. IMPORT_Preview lists every new, updated and unchanged row and every error, field by field. Nothing is written yet.
2. **Commit import.** A snapshot is taken, valid rows are written with history (source = the file name), rows with errors are skipped, and the files move to Imports\Done.

A template from another model, a file still open in Excel, or a type that no longer exists is refused at file level and left where it is.

Export. Choose all entities, one type, everything within 1 to 5 relationships of one entity, or the full model including the metamodel, as Excel or CSV. The file goes to Exports.

Example: the application owners send in a list. Generate a template for Application Component with pre-fill Y, send it out, put the returned file in Imports, preview, check the Update rows on IMPORT_Preview, and commit.

Reports

Three kinds of report serve three needs: EA_Reports.xlsx for everyday dashboards, Entity detail and Impact for questions about one entity, and Power BI for richer analysis.

EA_Reports.xlsx. Click **Refresh reports** on EA_Core's Home: it validates, saves EA_Core, then builds or refreshes EA_Reports.xlsx in the model folder. It has no macros and finds EA_Core in its own folder, so anyone can open it on its own and use Data > Refresh All. It reads the last saved EA_Core.

Page	Shows
Dashboard	Counts, errors, orphans, data-quality %, entities by domain and type, lifecycle, repository area
Entity list	Every entity, filtered with slicers such as type and domain
Relationships	Every relationship with both ends named
Matrix	A pivot of source type × target type for one relationship, such as applications × capabilities
Data quality	Every open issue from Validate All

Page	Shows
Orphans	Entities with no relationships

Entity detail and **Impact** (both Homes, or right-click rows) paint RPT_EntityDetail and RPT_Impact in EA_Entities. Detail shows attributes, relationships with working links, and recent changes. Impact lists everything within the hops you choose and how each entity is reached, for example "Claims Department, served by Claims Platform".

Power BI. Open Reports\EA_Reports.pbip in Power BI Desktop, set the ModelFolder parameter and refresh. The model is ready (entities, attributes in long form, types, relationships, issues, six measures); lay out the pages described in Reports\README_PowerBI.md, about 15 minutes. New attributes and types appear after a refresh without changing the model.

Diagrams with draw.io

Diagrams are drawn in draw.io Desktop and kept in step with the repository: every shape and connector carries the ID and type of the repository object it shows. Open the tools with **Diagrams (draw.io)** on either Home.

Load the libraries once. In draw.io: File > Open Library from > Device, pick Library\ArchiMate4.xml, then Library\ModelLibrary.xml. The *ArchiMate4* palette has the 42 element shapes in domain colours and, at the bottom, one connector per relationship. The *ModelLibrary* palette holds every active entity, titled "Type: Name (Display ID)".

Draw.

- New element: drag it from ArchiMate4 and type its name. For types with required attributes, select the shape, press Ctrl+M (Edit Data) and fill them in, for example Owner and Lifecycle for an Application Component.
- Existing element: drag it from ModelLibrary; it is the real repository entity.
- Relationship: drag a connector from the bottom of ArchiMate4 and attach both ends to shapes.

Generate from the repository. Pick content (around one entity with hops, one type, or all entities), a file name and a page name, then **Generate / add to diagram**. Shapes are placed in bands by domain; tidy the layout in draw.io. Generating into an existing page adds only what is missing and keeps your layout. Right-click rows > Diagram of selected entities does the same for a selection.

Harvest into the repository. Save and close the diagram in draw.io, then **Choose diagrams and preview**. HARVEST_Review lists:

Found in the diagram	Result
New shape from ArchiMate4	New entity; mandatory values come from Edit Data
New shape with the name of an existing entity of that type	Linked to that entity, no duplicate
Shape renamed on the diagram	Rename of the entity
New connector	New relationship, if ArchiMate allows it
Relationship ArchiMate does not allow	Error with the reason
Plain shape or note	Ignored
Shape or connector deleted	Nothing; deleting in draw.io never deletes in the repository

Commit harvest takes a snapshot, writes the changes with history, and stores the new IDs in the diagram file.

Keep diagrams current. Update diagrams from model refreshes names on every diagram and outlines retired or deleted entities in red. **Views register** lists every page with its status: OK, stale, relationships not yet drawn, or file missing.

Open and export. Open diagram starts draw.io on a file. **Export images** writes PNG, SVG or PDF of every page to Diagrams\Export\ for slides and reports.

If the Workbench says a page is saved compressed, untick File > Properties > Compressed in draw.io and save again.

Parent and child models

A child is a copy of part of a model for a project or team; its changes are merged back into the parent with a field-by-field review. Use **Parent and child models** on either Home.

1. **Create the child (in the parent).** Name it and pick a scope: around entities (Display IDs plus hops), one entity type, a repository area, or the whole model. The child is a new folder next to the parent, `EA_Model_<Parent>_<Child>`. Entities outside the scope that in-scope relationships point at come along as read-only references, so links still resolve. Diagrams come along when every shape on them is in the child.
2. **Close the parent, open the child.** Every model has an EA_Core.xlsm and Excel cannot open two workbooks with the same name, so close both parent workbooks first. Open the child's EA_Entities.xlsm: its first open removes everything outside the scope and saves the merge base.

3. **Work in the child** exactly as in any model.
4. **Prepare merge package (in the child).** It writes Merge\ChildPackage.xlsx. Close the child.
5. **Choose child and preview (in the parent).** Pick that package. MERGE_Review compares three versions of every record: the base (the child as it was created or last merged), the parent now and the child now.
6. **Decide and apply.** Set Keep on each row, then **Apply merge**. A snapshot is taken first; every change is logged with source "Merge <child>".

Base → parent → child	Row on MERGE_Review	Default Keep
Changed in the child only	Update	Child
Changed in the parent only	No row; the parent's value stays	—
Same change on both sides	No row	—
Different changes on both sides	Conflict	Parent
New in the child	Add (Display ID kept if free, else renumbered)	Child
Retired on one side, changed on the other	Conflict	Parent
Changed reference entity (outside the child's scope)	Info, not merged	Skip

Conflicts are per attribute, so a parent edit to Owner and a child edit to Lifecycle on the same entity merge cleanly. New metamodel types, attributes and rules from the child are merged first. Diagrams new in the child are copied; changed ones arrive as `<name>_child.drawio` next to the parent's.

The child can carry on after a merge; the next merge brings only what changed since. Merging runs from child to parent; to pick up later parent changes, create a new child.

Quick reference

I want to	Where
Add or change an entity	Type sheet in EA_Entities, or Home > Add / edit entity

I want to	Where
Add a relationship	Right-click a row > Add relationship, or click + and type in a relationship row
Retire an entity	Set Status to Retired
See what changed on an entity	Right-click a row > History of this entity
Check data quality	Home > Validate All, then SYS_Issues in EA_Core
Settle differences after a sync	SYNC_Review in EA_Core > Apply decisions
Add a type, attribute or rule	EA_Core Home > Add / edit type, Define rule, then Apply metamodel changes
Take or restore a snapshot	Home > Version history
Load many entities	Templates, import, export > Generate template, then Choose files and preview, Commit import
Export data	Templates, import, export > Export
Dashboards	EA_Core Home > Refresh reports, then EA_Reports.xlsx
One entity in detail, or its impact	Right-click rows > Detail of selected entities / Impact of this entity
Draw a diagram	Diagrams (draw.io) > Generate / add to diagram, or right-click rows > Diagram of selected entities
Bring a diagram into the repository	Diagrams (draw.io) > Choose diagrams and preview, Commit harvest
Give a project its own copy	Parent and child models > Create child
Merge a child back	Child: Prepare merge package. Parent: Choose child and preview, Apply merge